Cover Letter

Dear Editor,

We would like to submit our manuscript entitled "Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation" for consideration for publication in *Computers & Graphics*, as part of the Virtual Special Issue CAG_SS_CAD/Graphics 2026. The manuscript is original, has not been published previously, and is not under consideration elsewhere.

In this manuscript, we study a fine-grained inference-time control problem in geometry-controlled multi-view diffusion texturing: how strongly, and at which denoising stages, a geometric adapter should act. We first show that a uniform adapter scale is a blunt control — aggressive uniform scales improve structural metrics such as FG-SSIM but suppress the texture synthesis capability of the base model, causing surface flattening and loss of high-frequency detail. We then formalize the effect with a measurable Correction–Alignment–Interference (CAI) stage-utility model and derive Timestep-Conditioned Adapter Scaling (TCAS), a training-free low–high–low schedule that concentrates geometric correction in the middle denoising stage. The schedule C3 = (1.25, 2.50, 1.25) is selected using only a 24-object probe set and then evaluated unchanged, without further search, on the strictly disjoint 276-object holdout (pooled 300-object statistics also reported), where it matches the structure of aggressive scaling while improving PSNR by 0.96 dB, retaining substantially more texture, and receiving the highest overall preference (58.1%) in a blinded human study with cluster-aware significance analysis. To the best of our knowledge, no prior work analyzes the shape–texture trade-off of adapter scaling or derives a timestep-conditioned schedule from measurable stage utilities; the closest prior art (MVPainter, arXiv 2025; MV-Adapter, ICCV 2025; ControlNet, ICCV 2023; limited-interval classifier-free guidance, NeurIPS 2024) is cited and compared explicitly in Sections 2.3 and 4.4 of the manuscript.

The reference list has been carefully verified: citations that previously existed only as preprints have been updated to their peer-reviewed versions where available (e.g., TEXTure at SIGGRAPH 2023, Text2Tex at ICCV 2023, SyncDreamer at ICLR 2024, Wonder3D at CVPR 2024, Era3D at NeurIPS 2024, T2I-Adapter at AAAI 2024, DoRA at ICML 2024, MV-Adapter at ICCV 2025), genuinely unpublished works retain their arXiv identifiers, and recent relevant literature from *Computers & Graphics*, *IEEE TVCG*, *Computer Graphics Forum*, SIGGRAPH Asia, and *The Visual Computer* is now cited and discussed. For transparency, a copy of the original conference submission (CAD/Graphics 2026, Paper 75) is included as supplementary material.

Thank you very much for your attention and consideration.

Sincerely,

The Authors

(Per the journal's double-anonymized review policy, this cover letter contains no author details; author names, affiliations, and the corresponding author's contact information are provided on the separately uploaded title page and in Editorial Manager.)